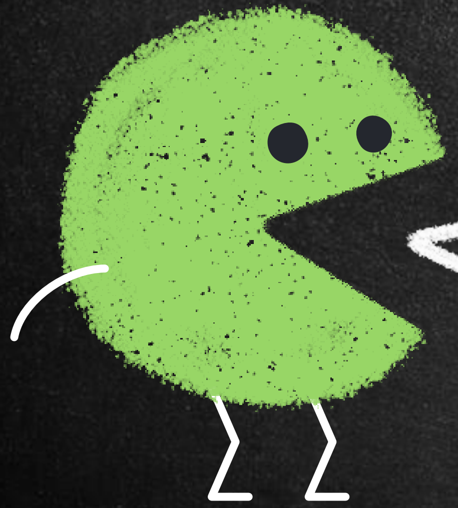
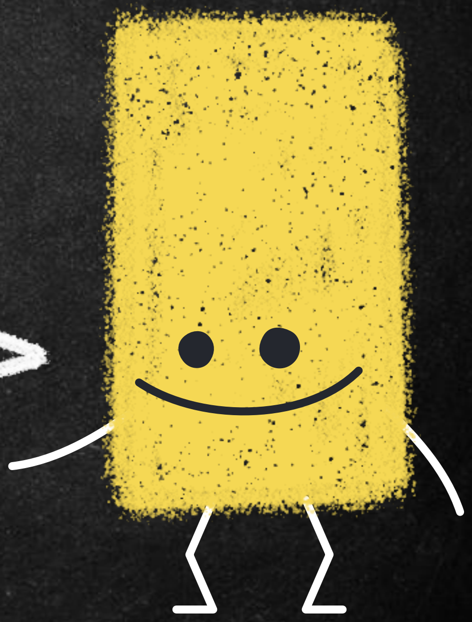


Digital Forensics
Unit – 2



Hello!

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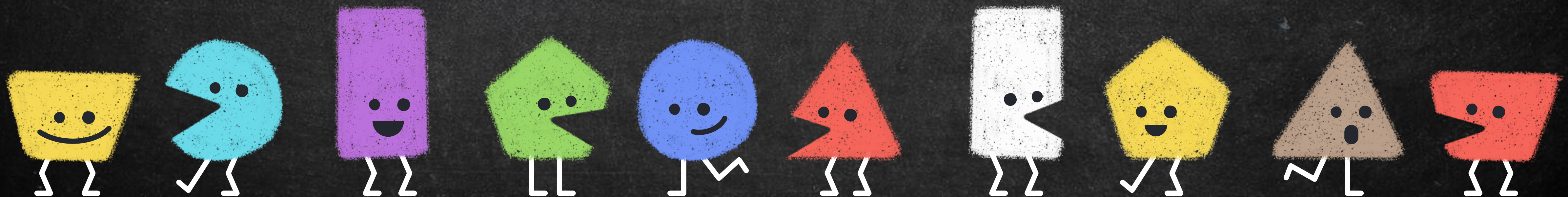
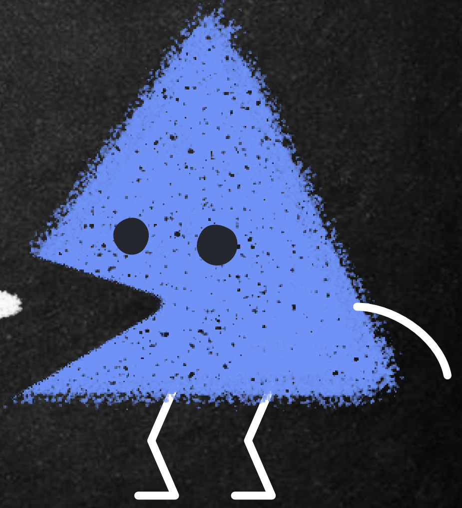


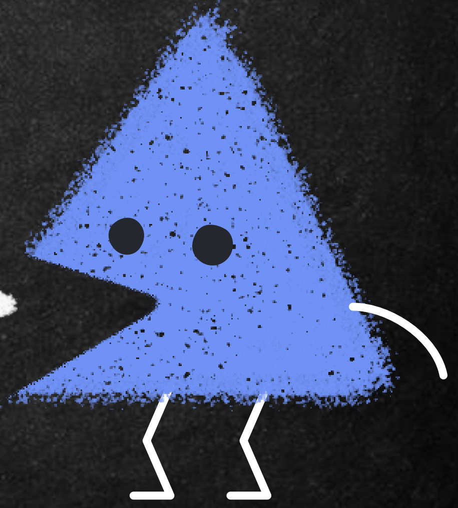
Basics

- → *The Boot Process*
  - *Kernel*
  - *BIOS*
  - *Bootstrapping*
- → *UEFI*
- → *MBR*
  - *Master Partition Table*
  - *Master Boot Code*
  - *Disk Signature*

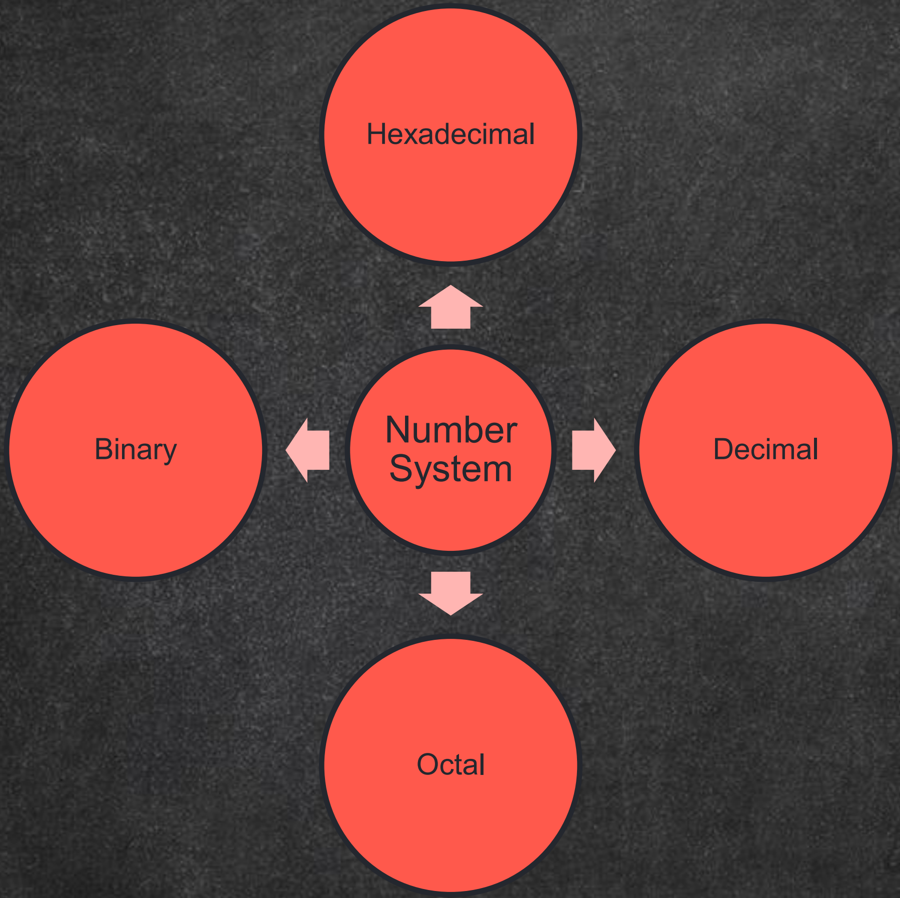


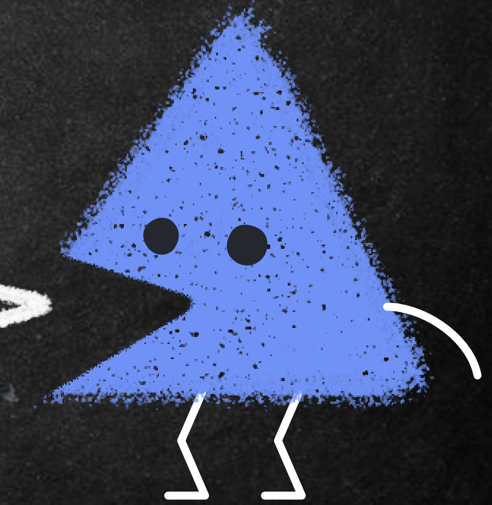
Number System

- → A number system is a method to represent (write) numbers.
- → Every number system has a set of unique characters or literals.
- → The count of these literals is called the radix or base of the number system.
- → Number systems are also called positional number system because the value of each symbol (i.e., digit and alphabet) in a number depends upon its position within the number.



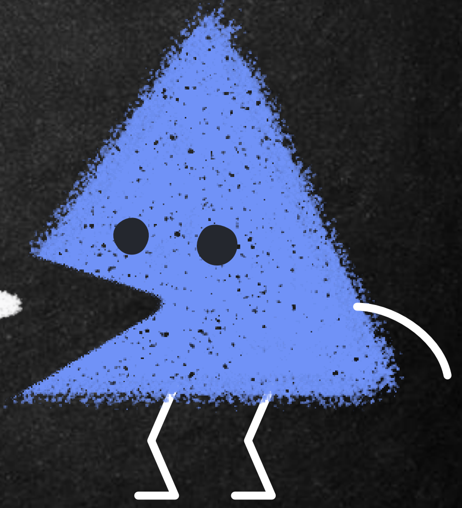
Number System





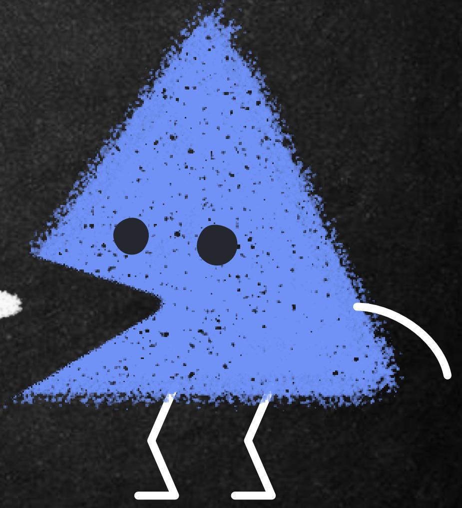
Conversion From Decimal

- Step 1: Divide the given number by the base value (b) of the number system in which it is to be converted
- Step 2: Note the remainder
- Step 3: Keep Dividing until the quotient is zero
- Step 4: Write the noted remainders in the reverse order (from bottom to top)



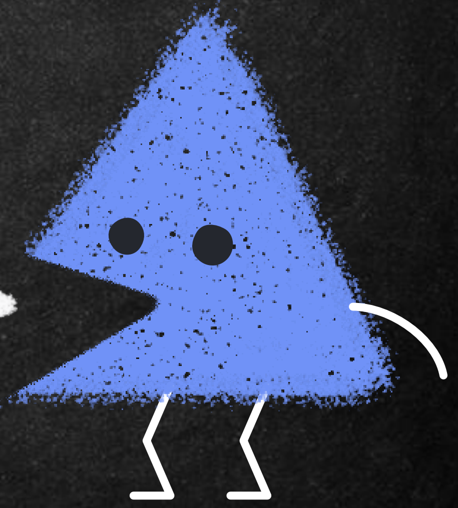
Conversion to Decimal

- Step 1: Write the position number for each alphanumeric symbol in the given number
- Step 2: Get positional value for each symbol by raising its position number to the base value b symbol in the given number
- Step 3: Multiply each digit with the respective positional value to get a decimal value
- Step 4: Add all these decimal values to get the equivalent decimal number

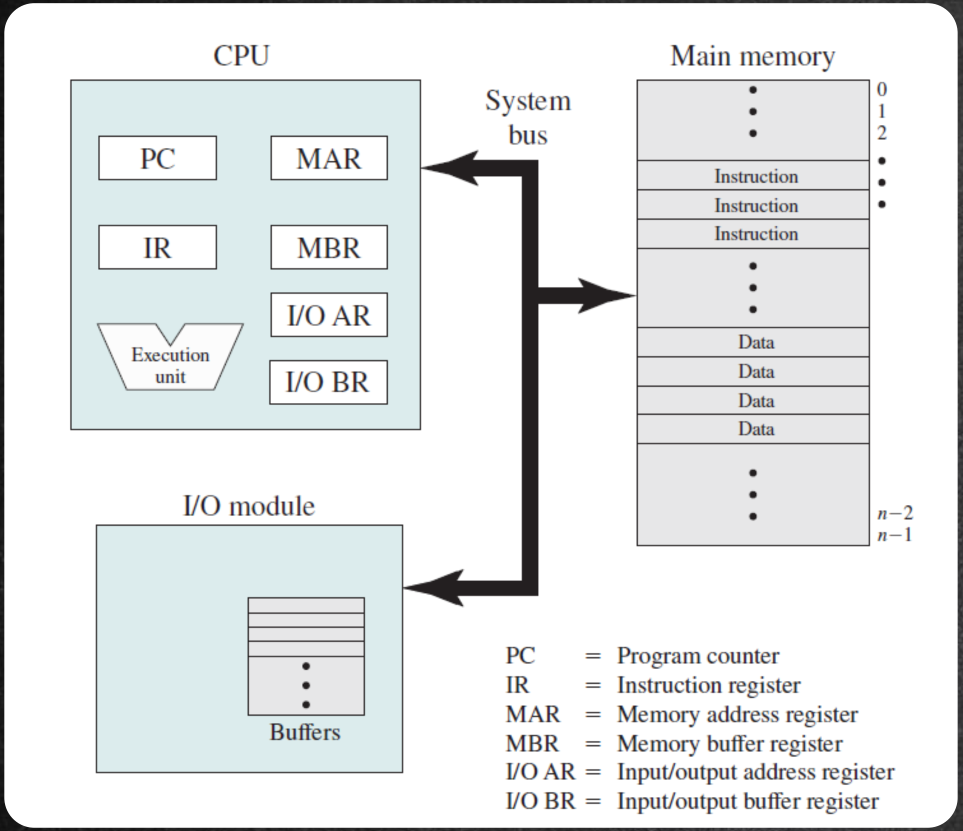


Task 1

- → Binary to Decimal
  - 100100111001
- → Hexadecimal to Decimal
  - A1B8C5
- → Octal to Decimal
  - 743924



Basic
Elements of Computer Systems



Processor

Main Memory

I/O Modules

System Bus

Overview: Computer Systems

Processor

Controls the operation of the computer and performs its data processing functions. When there is only one processor, it is often referred to as the central processing unit (CPU).

Main memory

Stores data and programs. This memory is typically volatile; that is, when the computer is shut down, the contents of the memory are lost. In contrast, the contents of disk memory are retained even when the computer system is shut down. Main memory is also referred to as real memory or primary memory.

Overview: Computer Systems

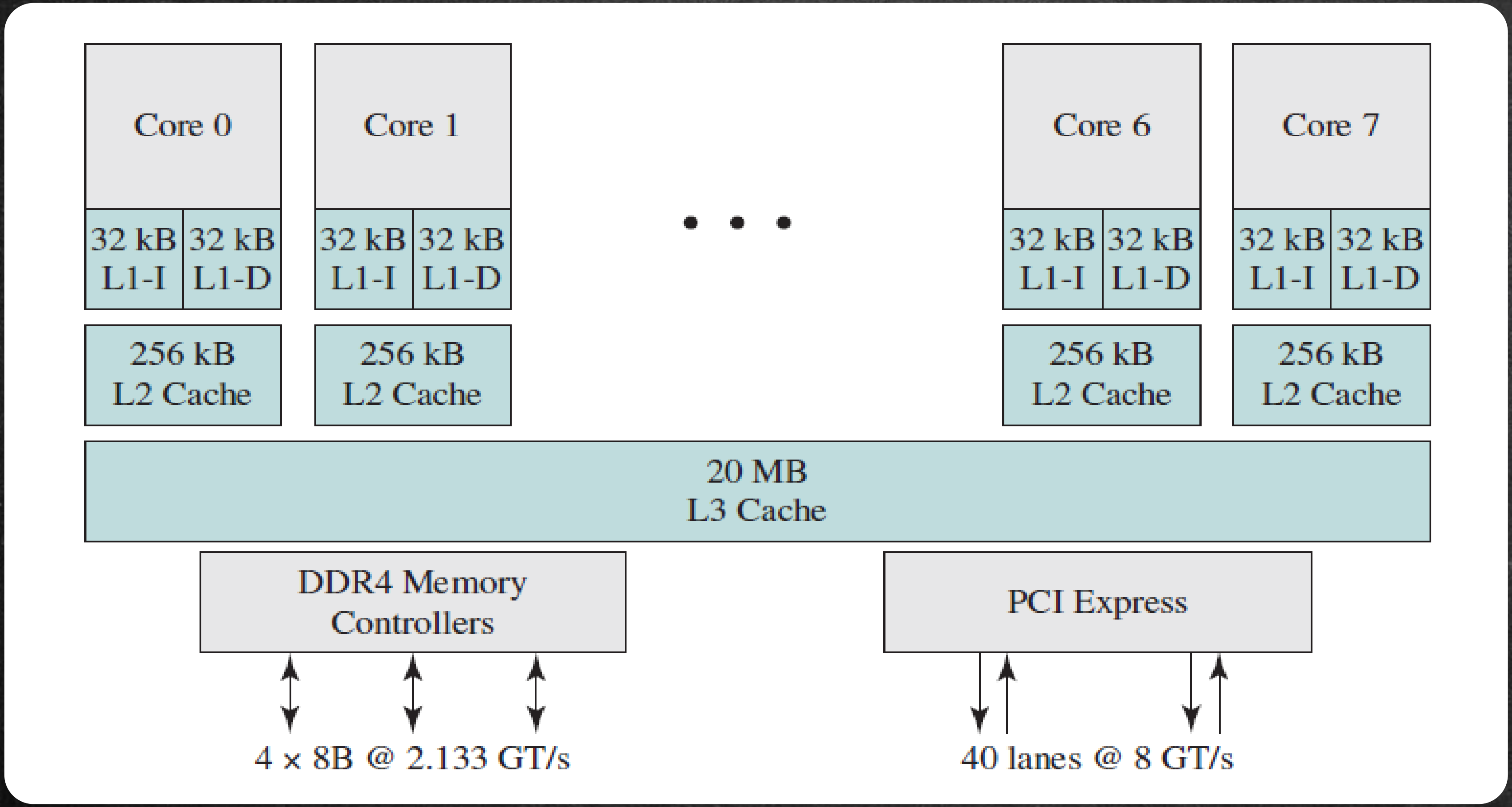
I/O Modules

Move data between the computer and its external environment.
The external environment consists of a variety of devices, including secondary memory devices (e.g., disks), communications equipment, and terminals.

System bus

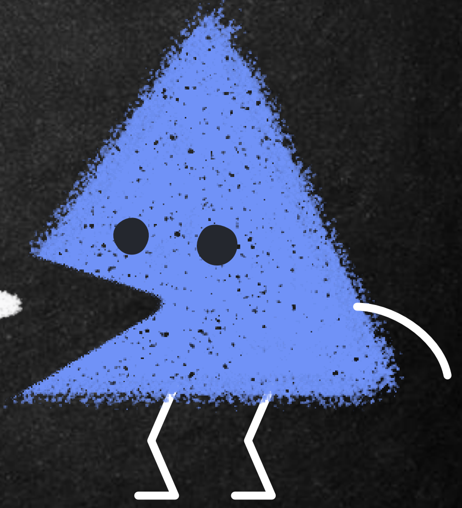
Provides for communication among processors, main memory, and I/O modules.

Chip Diagram of 8 Core Processor



Physical and Logical Storage

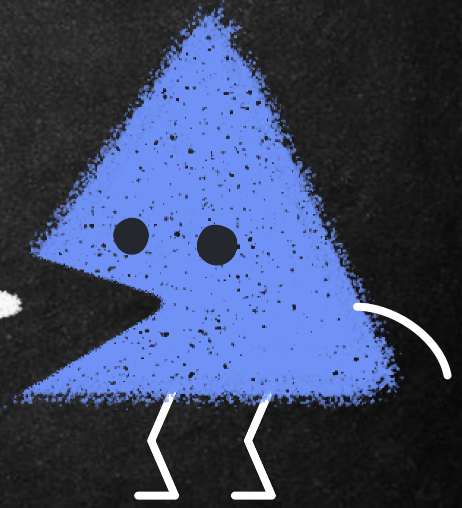
- → File storage and recording is largely controlled by the operating system.
- → A **byte** is comprised of 8 bits and is the smallest addressable unit in memory.
- → A **sector** on a magnetic hard disk represents 512 bytes, or 2048 bytes on optical discs.



Physical and Logical Storage

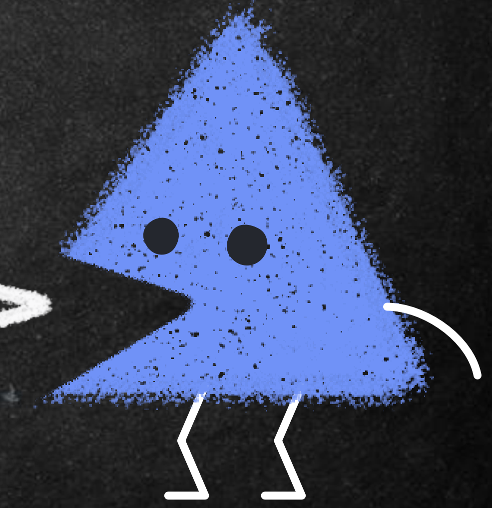
→ A **bad sector** is an area of the disk that can no longer be used to store data.

→ Bad sectors can be caused by viruses, corrupted boot records, physical disruptions, and a host of other disk errors.



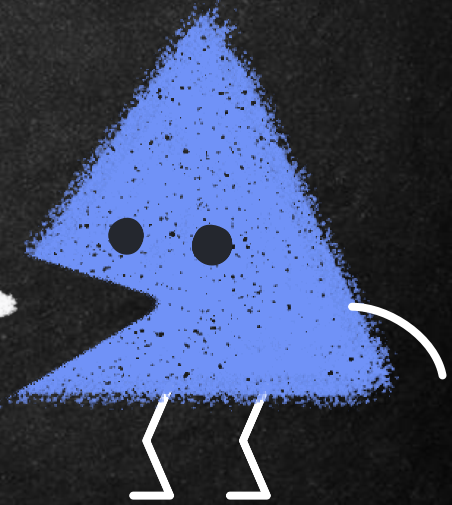
Physical and Logical Storage

- A **cluster** is a logical storage unit on a hard disk that contains contiguous sectors.
- When a disk volume is partitioned, the number of sectors in a cluster is defined.
- A cluster can contain 1 sector (512K) or even 128 sectors (65,536K).

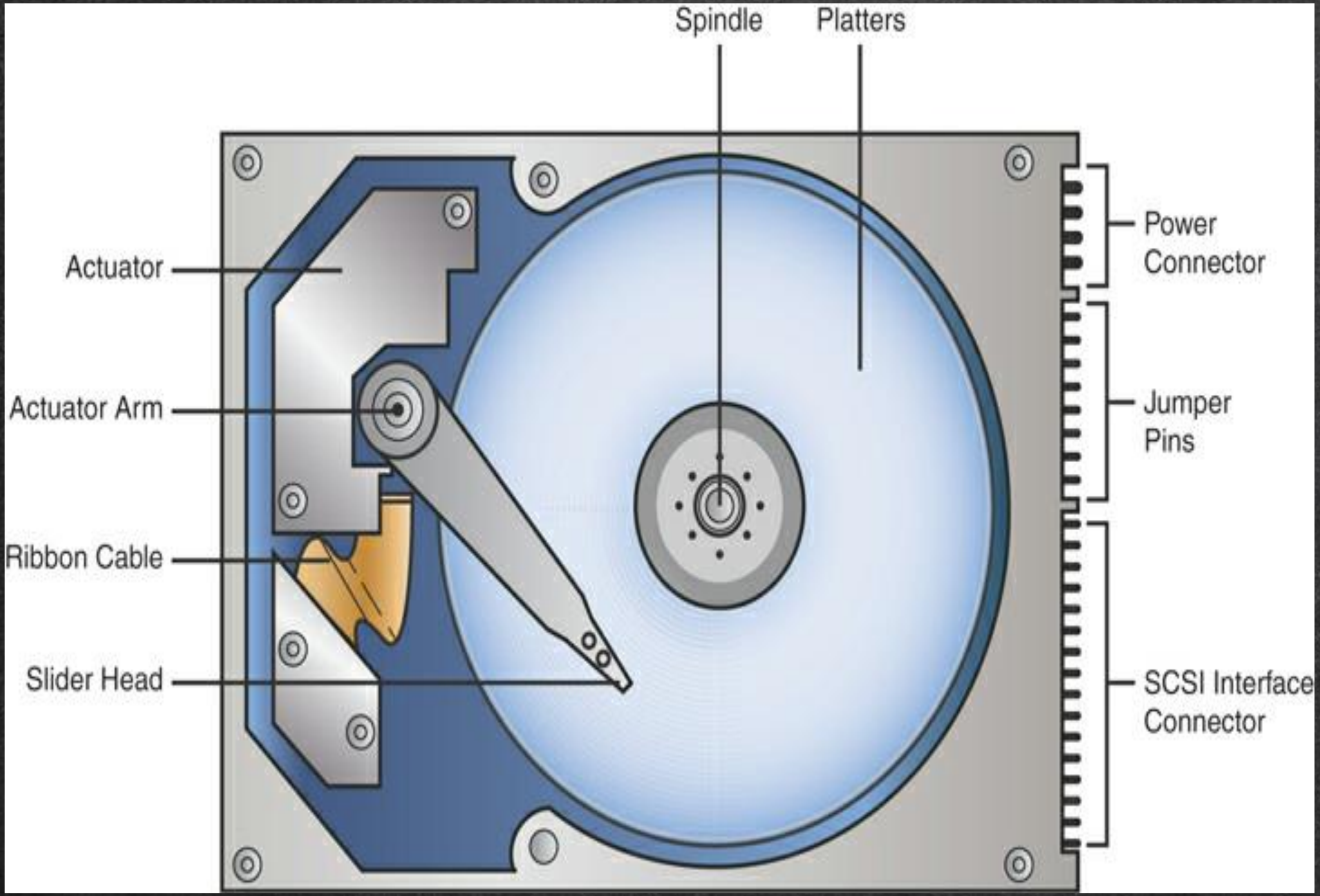


Physical and Logical Storage

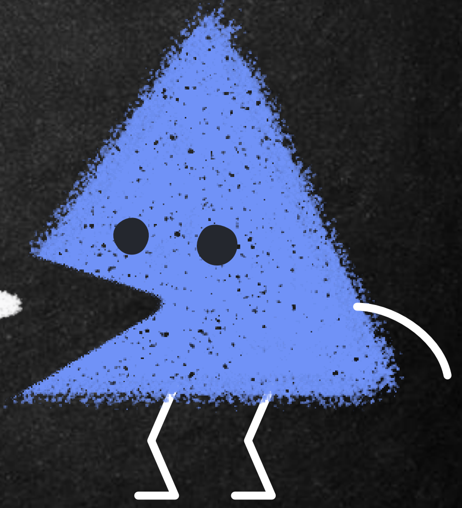
→ **Tracks** are thin, concentric bands on a disk that consist of sectors where data is stored.



The physical layout of a hard disk



The physical layout of a hard disk

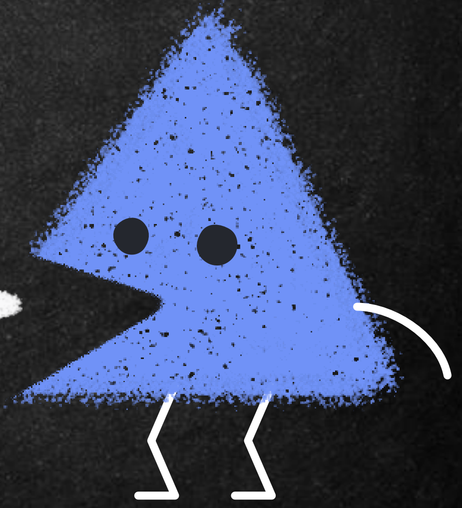


The physical layout of a hard disk

- A platter is a circular disk made from aluminum, ceramic, or glass that stores data magnetically. A hard drive contains one or more platters, and data can usually be stored on both sides of this rigid disk.

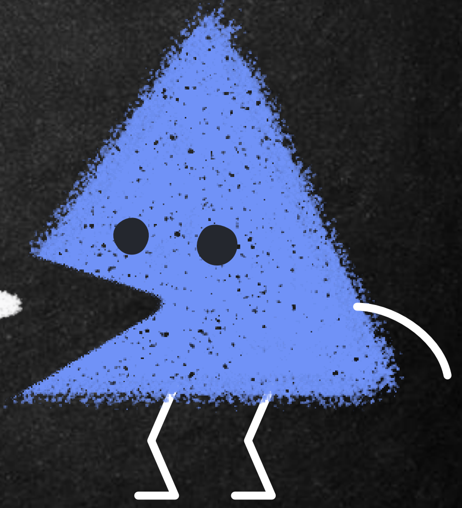
The physical layout of a hard disk

→ A spindle, at the center of the disk, is powered by a motor and is used to spin the platters. An arm sweeps across the rotating platter in an arc.



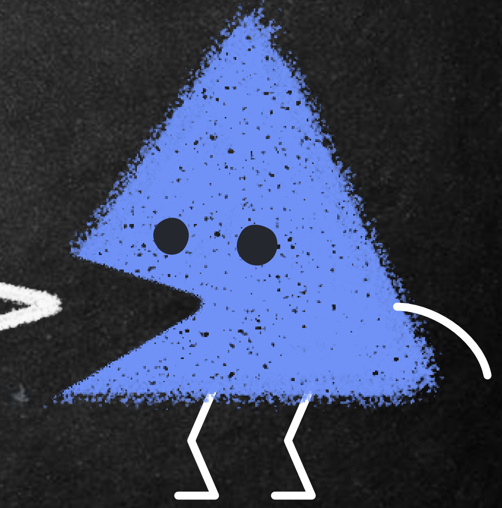
The physical layout of a hard disk

- → This actuator arm contains a read/write head that modifies the magnetization of the disk when writing to it.
- → There are generally two read/write heads for each platter because a platter usually contains data on both sides.



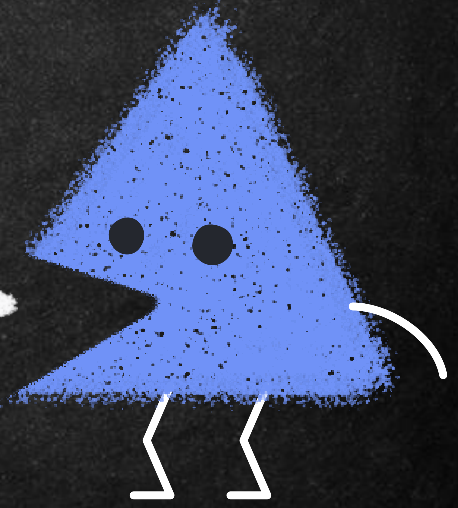
The physical layout of a hard disk

- The arm and head are nanometers from the platter. Therefore, hard drives must be handled very carefully; any impact on the hard drive could render the read/write head useless.
- Additionally, the examiner must be very careful not to let any magnetic device near a hard drive.



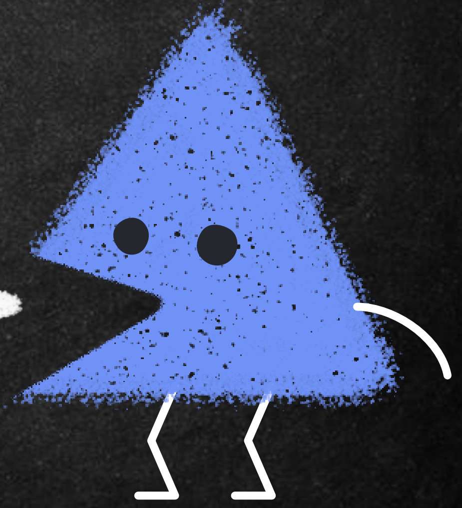
The physical layout of a hard disk

- A cylinder is the same track number on each platter, spanning all platters in a hard drive.
- Disk geometry refers to the structure of a hard disk in terms of platters, tracks, and sectors.

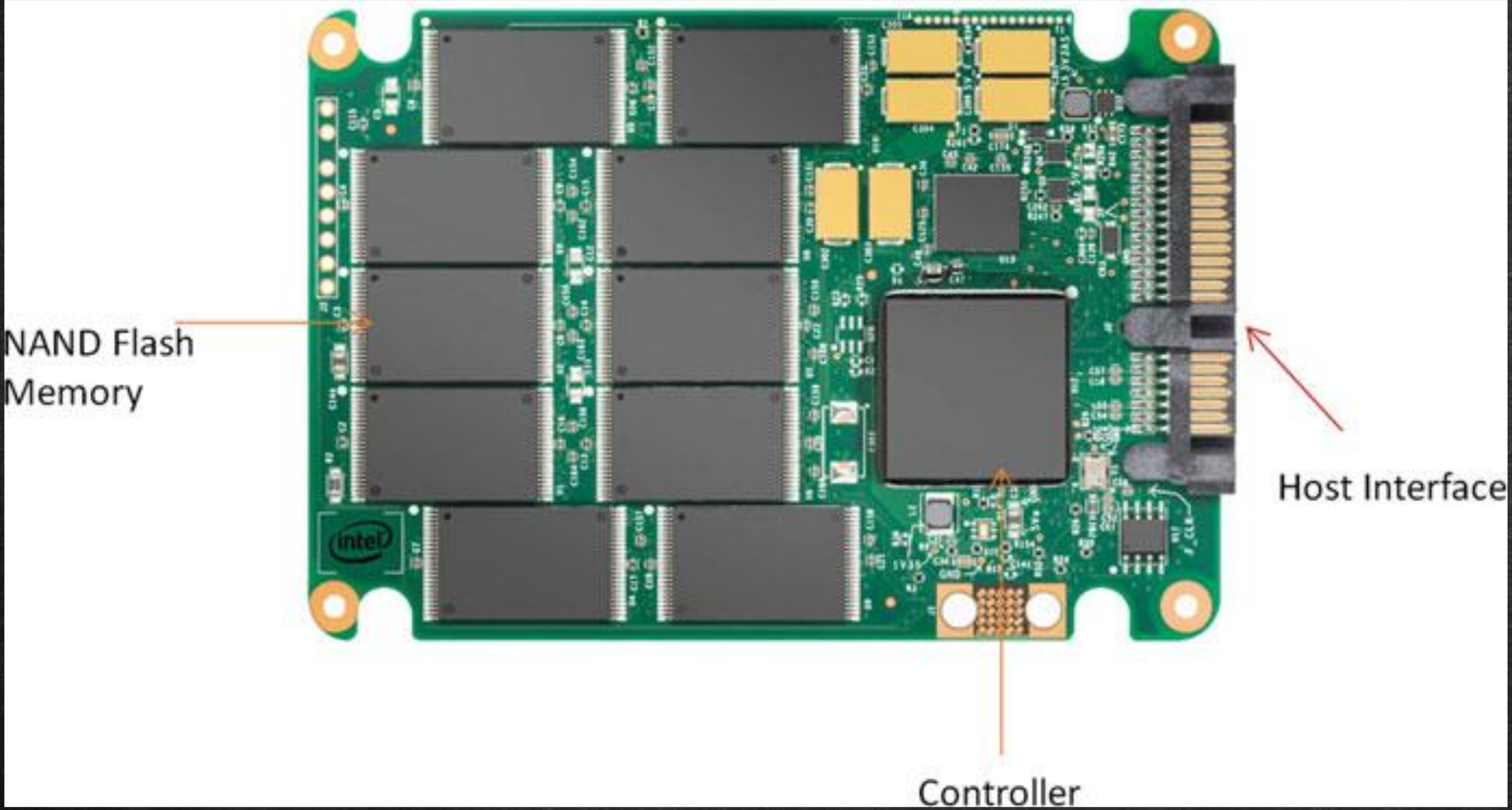


The physical layout of a hard disk

- The capacity of a hard disk drive can be calculated using the following formula:
- Number of cylinders × Number of heads × Number of sectors × Number of bytes per sector

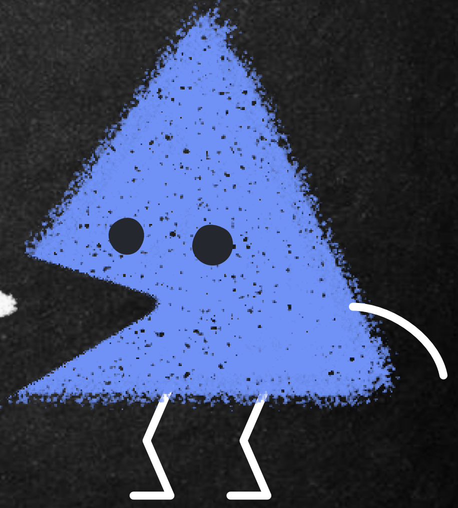


The physical layout of a Solid State Drive



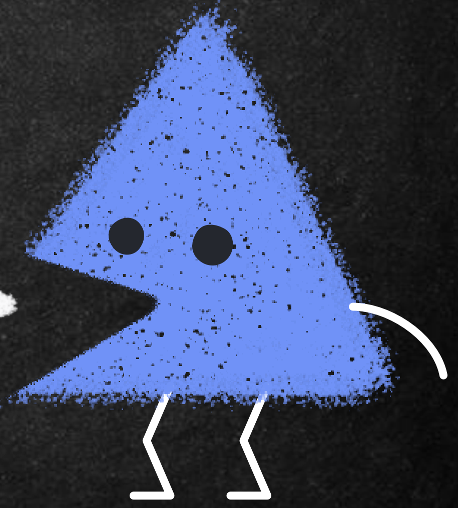
HPA Drive Areas

- *Host Protected Area (HPA)* was introduced in the ATA/ATAPI-4 standard and allowed system vendors to reserve portions of the disk for use outside the normal OS.
- For example, the HPA can be used to store persistent data, system recovery data, hibernation data, and so on.
- Access to this data is controlled by the system firmware rather than the installed OS.



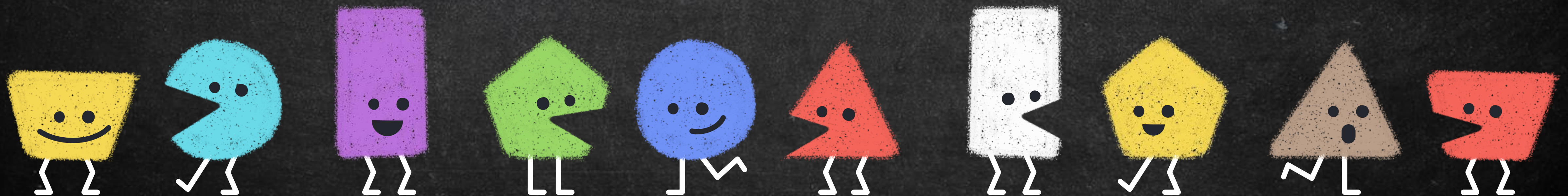
DCO Drive Areas

- *Device Configuration Overlay (DCO)* feature was introduced in the ATA/ATAPI-6 standard and provided the ability to control reported disk features and capacity.
- This allowed system vendors to ship disks from multiple manufacturers while maintaining identical numbers of user-accessible sectors and features across the disks.
- This facilitated easier support and drive replacement.
- Both the HPA and DCO can coexist; however, the DCO must be created first, followed by the HPA.
- The HPA and DCO have been misused by criminals and

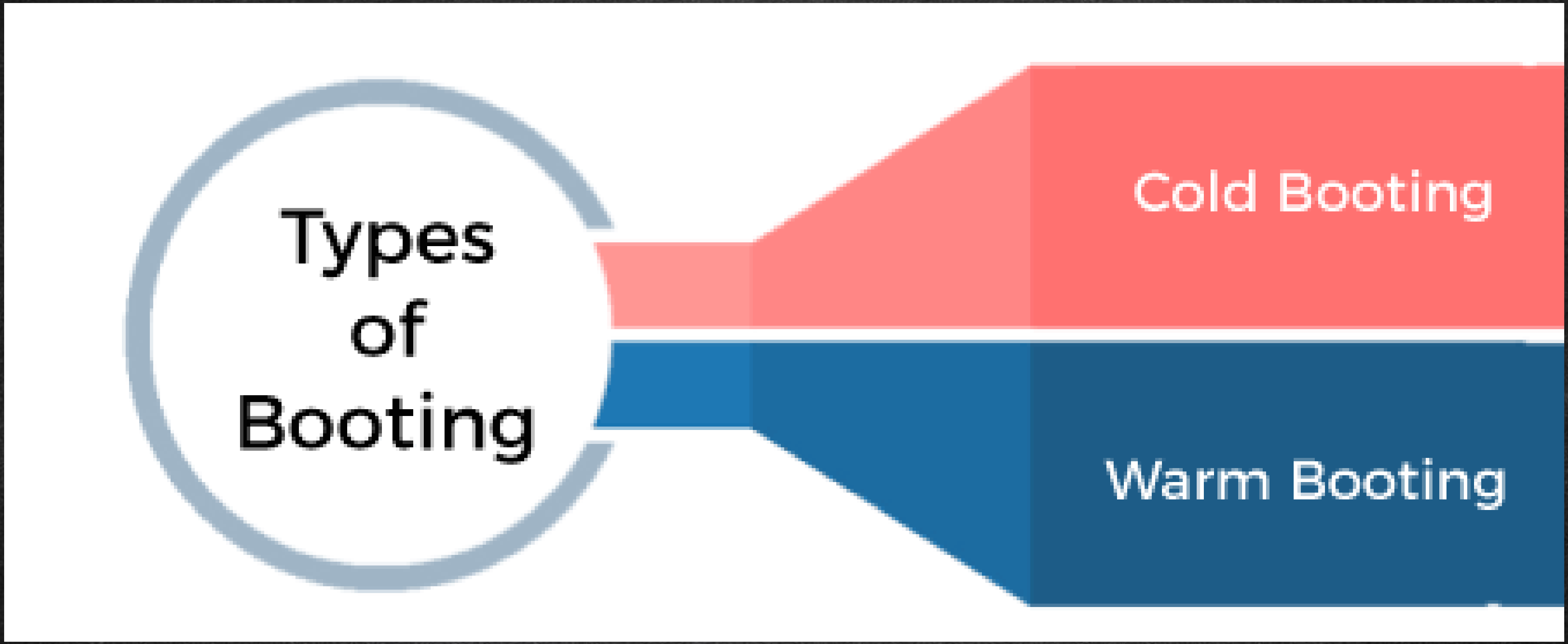


Booting Process

BIOS is loaded → Power-on self-test (POST) is completed → Operating system is loaded → System configuration is accomplished → System utilities are loaded → User is authenticated

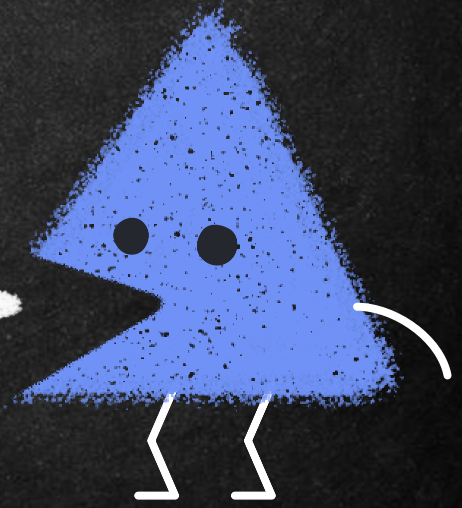


Booting Process



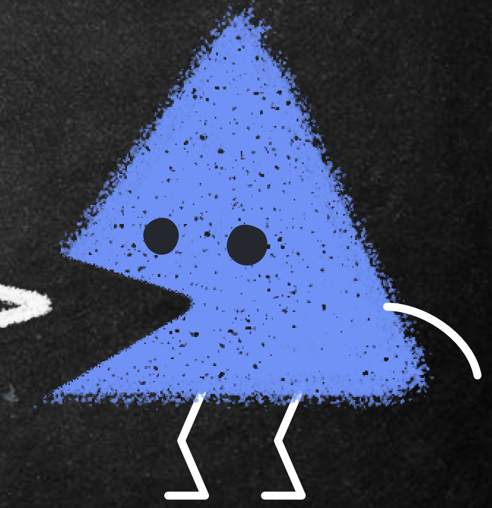
RAW Format

- → Raw images are not a format per se but a chunk of raw data imaged from an evidence source. Raw images contain no additional metadata aside from the information about the image file itself (name, size, timestamps, and other information in the image's own inode).
- → Extracting a raw image is technically straightforward: it is simply the transfer of a sequence of bytes from a source device to a destination file.
- → This is normally done without any transformation or translation. Disk block copying tools, such as dd and variants, are most commonly used to extract raw



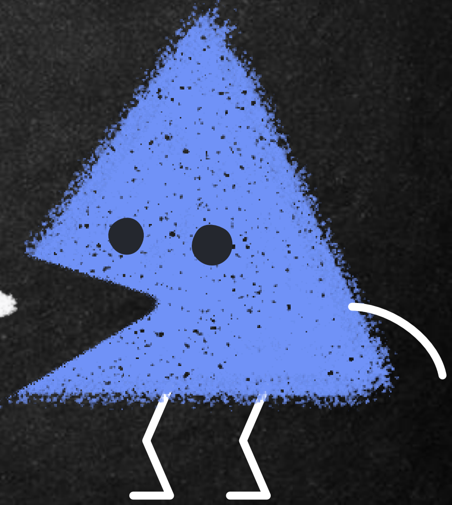
Forensic Format

- When imaging storage media as evidence, there is metadata about the investigation, the investigator, the drive details, logs/timestamps, cryptographic hashes, and so on. In addition to metadata, there is often a need to compress or encrypt an acquired image.
- Specialized forensic formats facilitate the implementation of these features.
- Forensic file formats are sometimes called *evidence containers*.



Locard’s exchange principle

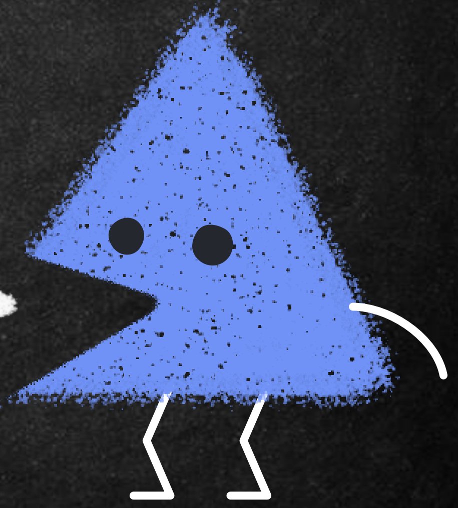
- Locard’s exchange principle says that in the physical world, when perpetrators enter or leave a crime scene, they will leave something behind and take something with them. Examples include DNA, latent prints, hair, and fibers.
- The same holds true in digital forensics. Registry keys and log files can serve as the digital equivalent to hair and fiber. Like DNA, our ability to detect and analyze these artifacts relies heavily on the technology available at the time.
- Viewing a device or incident through the “lens” of Locard’s principle can be very helpful in locating and



File Overview

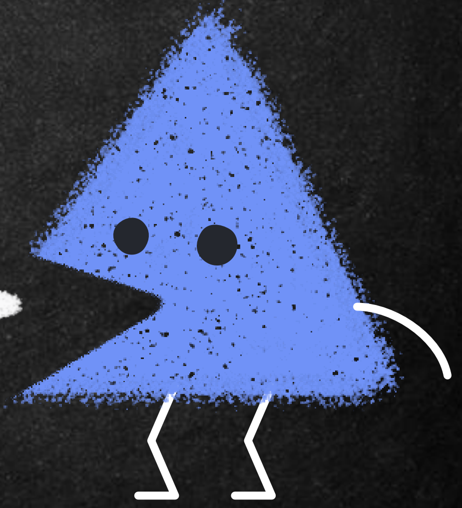
- files will have a specific structure or format. The start of the file is called the "header." The end of the file is known as the "footer." All of the bytes in between represent the remainder of the file.

- A file header is considered a form of metadata ("data about the data"). The header, like the extension, is used to identify the file type. However, unlike the extension, the header is much harder to change and is generally inaccessible to most users.



File Overview

- When the file is saved or written to the hard drive, it’s not necessarily saved to contiguous clusters
- These separate pieces of the file could be on different sides of a platter or on different platters altogether. When we double click on the file to open it, the computer gets the locations of all the sectors allocated to the file from the file system and recreates your file.
- To work on the file, it must be loaded into the computer’s main memory, also known as RAM. From here, the file is fed into the central processing unit (CPU) as we’re working with it.

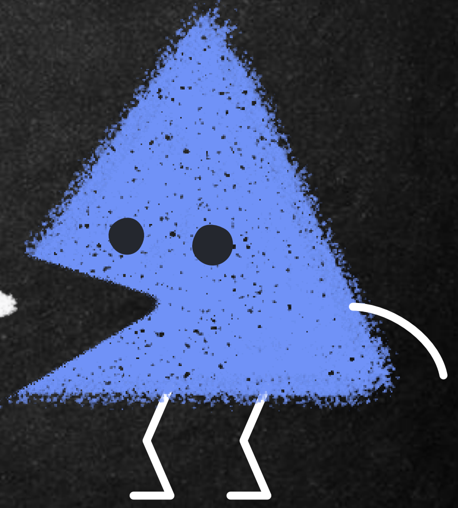


Data Type

- Data can be lumped into three broad categories: active, latent, and archival.

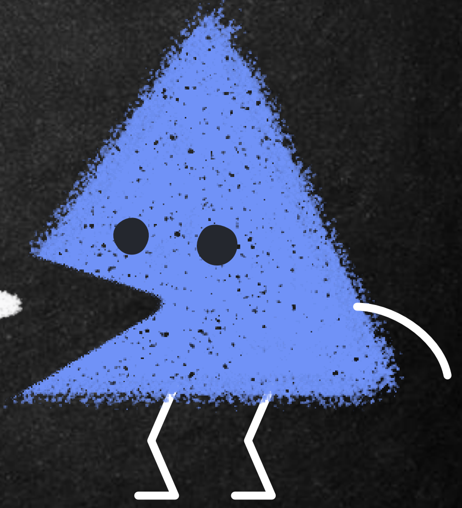
- Looking at data in this way helps in clarifying their location, how they’re accounted for by the file system, how they can be accessed by the user, and so on.

- It also helps to narrow down the cost and effort required to recover the data in question.



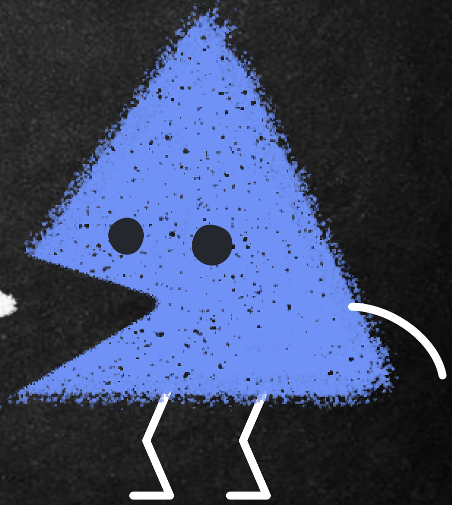
Active Data

- Active data are the data that we use every day on our computers. The operating system “sees” and tracks these files.
- These files can be located using Windows Explorer.
- These are the files that reside in the allocated space of the drive.
- These data can be acquired with standard forensic cloning techniques.



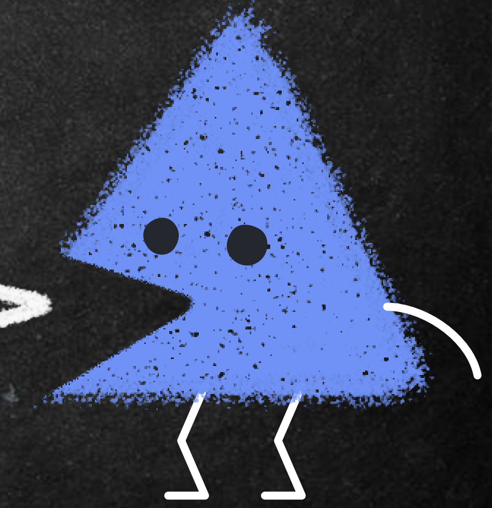
Latent Data

- Data that has been deleted or partially overwritten are classified as latent.
- These files are no longer tracked by the operating system and are therefore "invisible" to the average user.
- Looking for one of these files with Windows Explorer and it could nor be found.
- A bit stream or forensic image is required to collect these data.



Archival Data

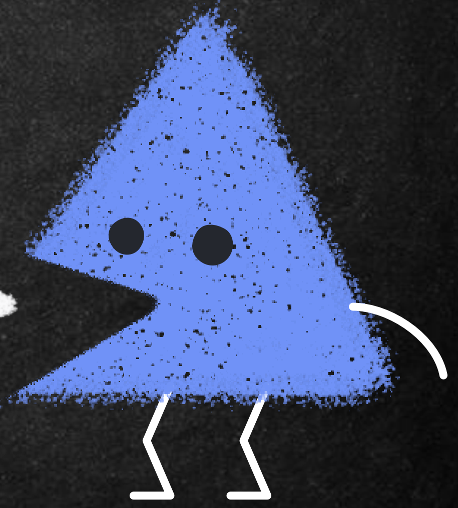
- Archival data, or backups, can take many forms.
- External hard drives, DVDs, and backup tapes etc.
- Acquisition of archival data can range from simple to extremely complex.
- The type and age of the backup media are major factors in determining the complexity of the process.



File System

→ With all the millions or billions of files floating around inside our computers, there has to be some way to keep things neat and tidy. This indispensable function is the responsibility of the file system.

→ The file system tracks the drive's free space as well as the location of each file. The free space, also known as unallocated space, is either empty or the file that previously occupied that locationhas been deleted.

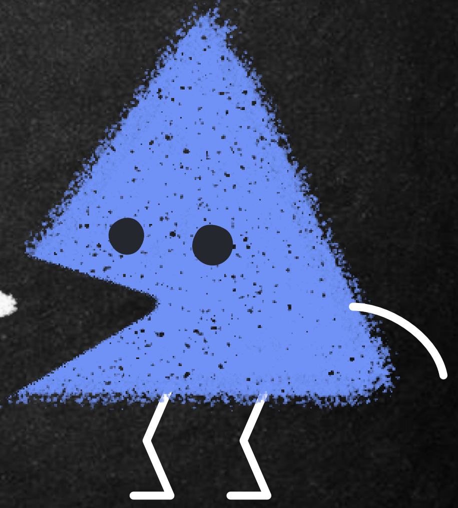


File System: FAT

→ File Allocation Table (FAT) is the oldest of the common files system.

→ It comes in four flavors: FAT12, FAT16, FAT32, and FATX.

→ Although not used in the latest operating systems, it can often be found in flash media and the like.

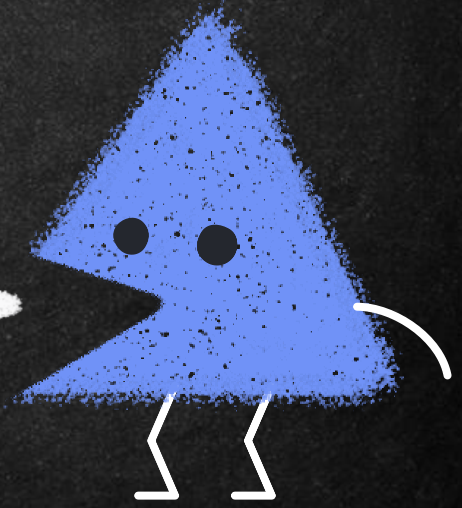


File System: NTFS

→ The New Technology File System (NTFS) is the system used currently by Windows 7, 8, 10, 11, Vista, XP, and Windows Server.

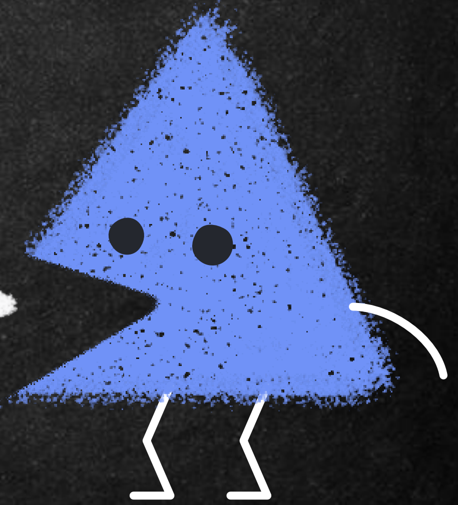
→ It's much more powerful than FAT and capable of performing many more functions.

→ For example, "NTFS can automatically recover some disk-related errors, which FAT32 cannot," it provides better support for larger hard drives, and better security through permissions and encryption.



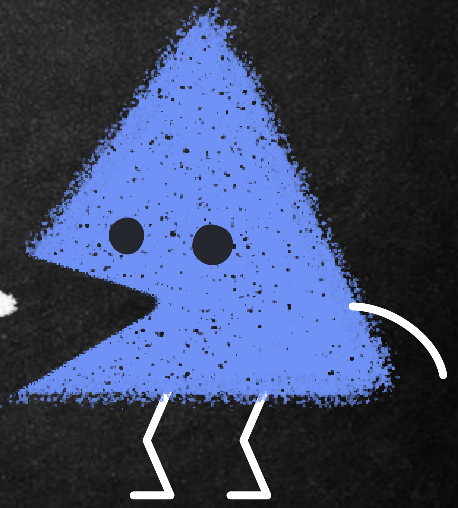
File System: HFS+

- Hierarchical File System (HFS+) and its relatives HFS and HFSX are used in Apple products.
- HFS+ is the upgraded successor to HFS.
- This newer version offers several improvements including improved use of disk space, cross-platform compatibility, and international-friendly file names



File System: Ext

- The extended file system, or ext, was implemented in April 1992 as the first file system created specifically for the Linux kernel.
- It has metadata structure inspired by traditional Unix filesystem principles, and was designed by Rémy Card to overcome certain limitations of the MINIX file system.
- There are other members in the extended file system family:
  - ext2, the second extended file system.
  - ext3, the third extended file system.

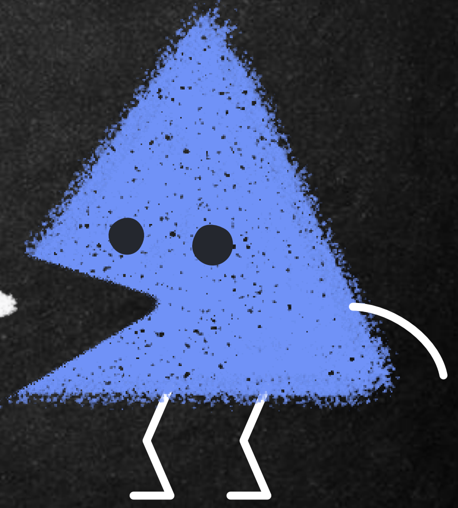


File System: APFS

→ Apple File System (APFS) is a proprietary file system developed and deployed by Apple Inc. for macOS Sierra (10.12.4) and later, iOS 10.3 and later, tv OS 10.2 and later, watch OS 3.2 and later, and all versions of iPad OS.

→ It aims to fix core problems of HFS+ (also called Mac OS Extended), APFS's predecessor on these operating systems.

→ APFS is optimized for solid-state drive storage and supports encryption, snapshots, and increased data



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- Practical Forensic Imaging: Securing Digital Evidence with Linux Tools by Bruce Nikkel
- A Practical Guide to Digital Forensics Investigations 2nd Edition by Darren R. Hayes